NUMBER THEORY ON ONE PAGE

Primes, the sieve and divisibility tests to 13

The sieve needs only primes up to the square root, and one ansatz explains the tests for 3, 7, 9, 11 and 13.

1 / THE RULE AND THE PICTURE

A composite number has a prime factor at most its square root, so the sieve stops there

1	**2**	**3**	4	**5**	6	**7**	8	9	10
11	12	**13**	14	15	16	**17**	18	**19**	20
21	22	**23**	24	25	26	27	28	**29**	30
31	32	33	34	35	36	**37**	38	39	40
41	42	**43**	44	45	46	**47**	48	49	50
51	52	**53**	54	55	56	57	58	**59**	60
61	62	63	64	65	66	**67**	68	69	70
71	72	**73**	74	75	76	77	78	**79**	80
81	82	**83**	84	85	86	87	88	**89**	90
91	92	93	94	95	96	**97**	98	99	100

prime · 2 · 3 · 5 · 7

colours show the smallest prime factor

If $N = pq$ with $p \le q$ then $p^2 \le N$, so a composite N has a prime factor $p \le \sqrt{N}$:

$$N \text{ is prime} \iff \text{no prime } p \le \sqrt{N} \text{ divides } N$$

Eratosthenes strikes out the multiples of each prime p, beginning at p^2 because the smaller multiples $2p, 3p, \dots$ were struck by smaller primes. The primes $2, 3, 5, 7$ clear everything up to 100 and leave its 25 primes, since the next prime 11 first acts at $11^2 = 121$. To settle every number below $17^2 = 289$ we need the primes up to 13, which leave 61 primes, and the tests below decide divisibility by each of them quickly.

prime p	2	3	5	7	11	13
first strike p^2	4	9	25	49	121	169
primes up to p settle $N <$	9	25	49	121	169	289

2 / THE ANSATZ / MULTIPLY BY THE INVERSE OF 10

Write $n = 10a + b$ and multiply by m with $10m \equiv 1 \pmod p$, so that $mn \equiv a + mb$

Because m is invertible modulo p, $p \mid n$ exactly when $p \mid a + mb$, which shortens n by one digit each time. The primes 2 and 5 divide 10 itself and use the last digit.

p	m with $10m \equiv 1$	**test on $n = 10a + b$**	**why**	**example**
2	none	last digit even	10 is even	2024
3, 9	1	digit sum, $a + b$	$10 \equiv 1$	$2025 \to 9$
4	none	last two digits	$4 \mid 100$	$2024 \to 24$
5	none	last digit 0 or 5	$5 \mid 10$	2015
7	5 or -2	$a - 2b$ (or $a + 5b$)	$10(a - 2b) + 21b = n$	2023
11	-1	$a - b$	$11a + (b - a) = n$	2024
13	4	$a + 4b$	$4n = 39a + (a + 4b)$	2015

3 / WHY THE TEST FOR 7 WORKS / $n = 2023$

Since $10a + b = 10(a - 2b) + 21b$ and $7 \mid 21b$, the number n is a multiple of 7 exactly when $a - 2b$ is

Subtracting twice the last digit from the rest shortens the number, and n differs from $10(a - 2b)$ by a multiple of 7 while 10 is invertible modulo 7. For 2023 we have $a = 202$ and $b = 3$, so $202 - 6 = 196$, then $19 - 12 = 7$, and 7 divides 7, so $7 \mid 2023$ and indeed $2023 = 7 \times 17^2$. The multiplier 5 gives the second test, since $5n = 49a + (a + 5b)$ and $7 \mid 49a$: $202 + 15 = 217 = 7 \times 31$.

4 / WHY THE TEST FOR 11 WORKS / $n = 2024$

Since $10 \equiv -1 \pmod{11}$, the number $n = 10a + b$ is $b - a$ modulo 11

Writing $10a + b = 11a + (b - a)$ shows that $11 \mid n$ exactly when $11 \mid a - b$. For 2024 we have $a = 202$ and $b = 4$, so $202 - 4 = 198 = 11 \times 18$, and $2024 = 8 \times 11 \times 23$. Applying the same step to every digit gives the alternating sum $4 - 2 + 0 - 2 = 0$, because $10^k \equiv (-1)^k \pmod{11}$, so the digits alternate in sign from the right.

5 / ONE TEST FOR 7, 11 AND 13 / THE NUMBER 1001

As $1001 = 7 \times 11 \times 13$ and $1000 \equiv -1$, alternating three-digit blocks decide all three

n	blocks	alternating sum	divisible by	factorisation
2023	$2 \mid 023$	$23 - 2 = 21 = 3 \cdot 7$	7	$7 \cdot 17^2$
2024	$2 \mid 024$	$24 - 2 = 22 = 2 \cdot 11$	11	$2^3 \cdot 11 \cdot 23$
2015	$2 \mid 015$	$15 - 2 = 13$	13	$5 \cdot 13 \cdot 31$

Powers of 10 modulo 7 run $1, 3, 2, 6, 4, 5$, modulo 13 they run $1, 10, 9, 12, 3, 4$, and modulo 11 they alternate $1, -1$; in both 7 and 13 the third power is -1, which is where the blocks of three come from.

ASIDE / THE NEXT PRIME

For 17 the multiplier is $m = 12$ and the test is $a + 12b \equiv a - 5b$, found the same way.

CHECK IN PYTHON

All three reductions agree with `n % p == 0` for every $n < 100\,000$.

Convention. $n = 10a + b$ with b the last digit; $\equiv$ is congruence; $\pi(100) = 25$ and 61 primes lie below $289 = 17^2$.

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